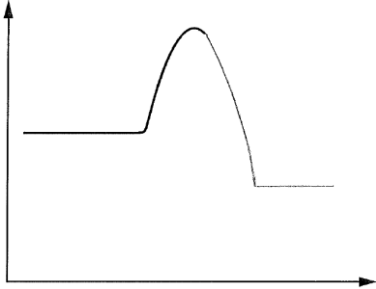


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		15		1		1	1	1
		(ii)		both reactions and reason needed A and B temperature increases / they get hotter / positive temperature change neutral answer – energy / heat given out		1		1		1
	(b)			 energy of products must be lower than energy of reactants accept without the horizontal line for products accept with energy of products on x-axis	1			1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			2 Al + 3 ZnCl₂ → 2 AlCl₃ + 3 Zn award (1) for product award (1) for balancing zinc atoms (even if aluminium chloride formula is incorrect)		2		2		
				Question 4 total	1	4	0	5	1	2